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Case Report

Barium sulfate aspiration: Severe chemical pneumonia induced by a massive reflux of contrast medium during small bowel barium enema

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ABSTRACT

Barium contrast radiography is a conventional procedure aimed at revealing lesions of the alimentary tract using barium sulfate on X-ray irradiation. Although it is widely used in clinics, adverse effects and complications are observed, such as anaphylaxis, granuloma, fecalithes, abdomen-leaking, embolism, bacterial contamination, and aspiration. We report a case of death due to a massive barium sulfate aspiration resulted from an air-barium double contrast enema radiography. A 25-year-old female patient was hospitalized with symptoms of abdominal distention, nausea, vomiting and diarrhea for three days. A progressive respiratory distress presented only 1 h after a small bowel air-barium double contrast enema. The patient died 11 h later. The result of autopsy revealed the cause of death to be severe chemical pneumonitis induced by gastric fluid which was aspirated into her lungs. Barium sulfate is generally recognized to be chemically inert for the respiratory system, but a mixture of barium sulfate with gastric contents is fatal. Here we intend to suggest that, when determining the potential cause of death, medical examiners should consider a patient's status quo as well as the possible adverse effects and complications caused by the barium sulfate preparation during gastrointestinal radiography.

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1. Introduction

Gastrointestinal contrast is a routine examination for medical diagnosis. It has been broadly used in hospitals. The procedure is usually performed with barium sulfate contrast, swallowed or administrated into the esophagus, stomach and duodenum for radiographic development. Barium sulfate is insoluble in water and forms a suspension. It is relatively safe in alimentary radiography because it is not a chemical irritant. Sellink [1] and Herlinger [2] reported in 1974 and 1978, respectively, that the extensive applications of barium ingestion for X-ray visualization can cause adverse effects and complications, such as anaphylaxis, granuloma, fecalithes, abdomen-leaking, embolism, bacterial contamination, aspiration, and even death [3,4]. However, it has not been fully recognized by clinicians and medical examiners [5]. Pulmonary barium aspiration during the examination is not unusual, but fatal cases are rare and they are mostly confined to a

radiographic diagnosis [4,6,7]. Tsokos et al. [8,9] reported a case of fatal barium sulfate aspiration with a moderate dose (100 ml) of gastric administration in barium radiography and the patient with gastric cancer died 35 h later.

Here, we report a case of a female patient who died of a massive barium sulfate aspiration due to a small bowel air-barium double contrast enema.

2. Case report

A 25-year-old female patient was admitted to hospital for abdominal distention, nausea, vomiting and diarrhea for three days. She was diagnosed with systemic lupus erythematosus (SLE) 2 years ago and had a medication history of prednisone. Physical examination showed the patient with abdominal distention, shifting dullness and tenderness of the umbilicus and hypogastrium. Intestinal X-ray suggested a small bowel obstruction (SBO). A small bowel air-barium double contrast enema was performed to define the lesion. Local anesthesia by lidocaine spray on the pharynx was performed and an orogastric tube was inserted down to the jejuno-duodenal junction. The procedure lasted for 30 min and approximately 1000 ml of barium sulfate contrast was administrated. The examination reported an inflammatory lesion. One hour after the procedure, the patient complained of

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Fig. 1. X-ray chest film shows multi-nodular and patchy consolidation.

palpitations, chest tightness and dyspnea. Her blood pressure (BP) dropped to 90/50 mmHg and the heart rate (HR) increased to 120 bpm. Labored breathing and fine crepitations were detected, especially in the left lung. In 3 h, her situation became worse with a temperature of 39.5 °C and HR of 176 bpm. Mild to moderate moist rale was obvious and an X-ray chest film suggested inhalation and chemical pneumonia (Fig. 1). Tachypnea and cyanosis of her face and lips were apparent and pressure of oxygen was 4.75 kpa. The patient died 11 h after the procedure.

Forensic necropsy was performed 48 h after her death. External examination indicated the patient to be a physically well adult female without any signs of mechanical injury or asphyxia. Internal examination revealed white contrast agents in esophagus, stomach and intestinal canals. An incomplete volvulus of the ascending colon was noted with distinct tissue edema 16 cm away from the appendix. Diffuse inflammatory infiltration was demonstrated microscopically.

Larynx and pulmonary edema was notable while the trachea, main and lobar bronchi were stained with contrast agents (Fig. 2). A 16 cm × 8 cm area filled with a certain amount of white plaques presented in the inferior lobe of the left lung; and a 15 cm × 7 cm area was presented in the right lung as well as the cut sections (Fig. 3). Significant inflammatory cells full of neutrophilic granulocytes were infiltrated in most of the alveolar cavities. Yellowish-brown depositions were diffusely found in alveolar

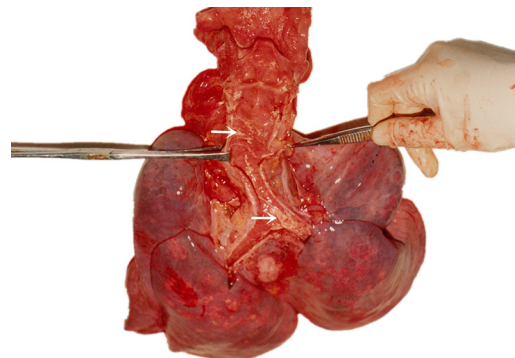


Fig. 2. White discoloration of the trachea and bronchi.

cavities and phagocytosed by macrophages. A fibrinous exudate, fibrinoid necrosis and hyaline membrane formation were observed (Fig. 4). A photo of a stained smear of barium sulfate contrast material is also added (Fig. 5).

The cause of death was determined as severe chemical pneumonia due to gastric fluid and barium sulfate aspiration of barium enema examination for radiographic purpose.

3. Discussion

Barium-containing contrast medium is widely used in gastrointestinal radiography for inflammation, tumor, polypus, diverticulum, varicosity, hernia, fistula, prolapse, volvulus and obstruction. Use of contrast agents enhances visualization of internal structures, and these agents are believed to be bland, defined, feasible and economical materials. Clinically, barium sulfate preparation for oral usage and enema is composed of barium sulfate and additives. The formulation can be tablets, powder, paste or suspension. Because of the variations in enteric mucosa, it is important to justify the stabilization, density, viscosity and chemical composition to achieve a desired result. Currently, nearly a hundred of additives are used in the barium contrast [10]. The application of a specific barium sulfate suspension depends on the target areas in clinics. Despite of little irritation, some of the additives, for instance, deflocculating agent such as sodium citrate, may still induce adverse effects as anaphylaxis or gastrointestinal hemorrhage.

Adverse effects and complications related to barium-containing contrast radiography of the upper gastrointestinal tract occur rarely [11] and most symptoms are mild and self-limited. However, rare complications such as aspiration should be given adequate attention. This complication is found predominantly in infant, old people, sedative takers, and patients with esophageal obstruction, respiratory disorder, dizziness, consciousness

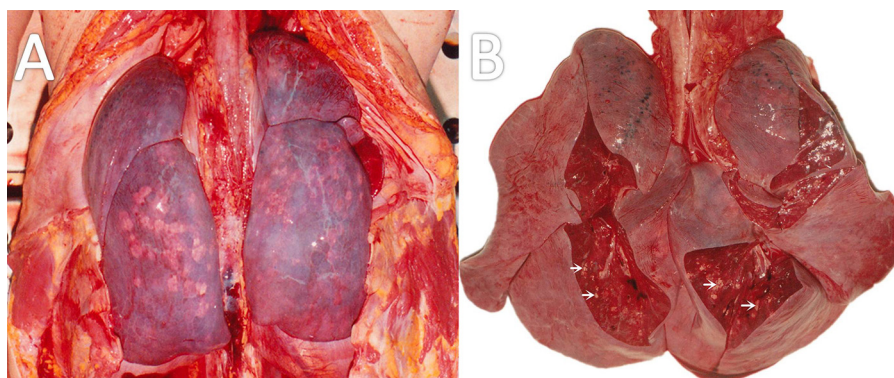


Fig. 3. A certain amount of white plaques presented in the lung (A) and the cut sections (B).

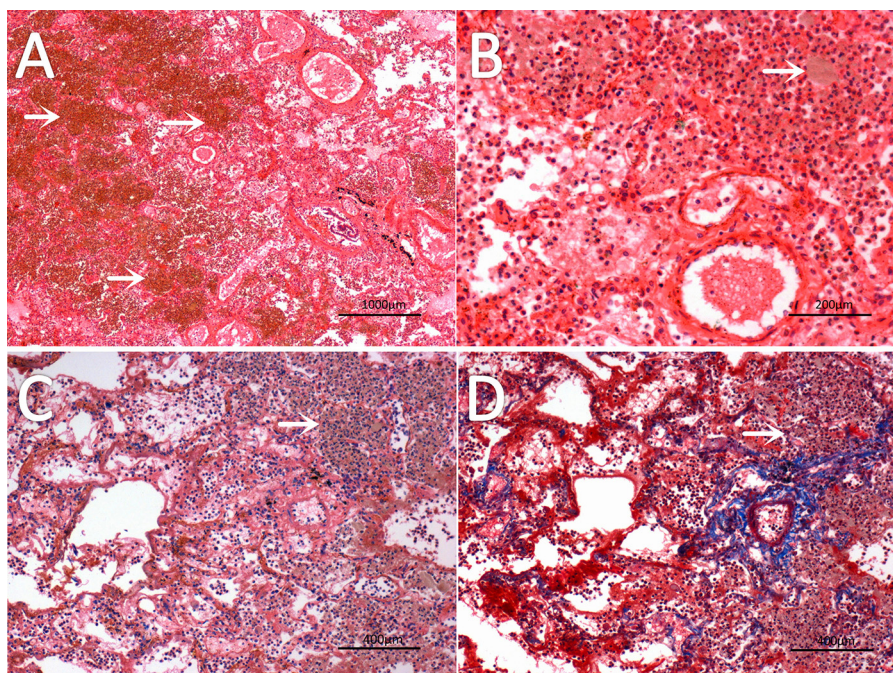


Fig. 4. Inflammatory cells such as neutrophil granulocytes were infiltrated in most of the alveolar cavities. Yellowish-brown barium contrasts were diffusely found in alveolar cavities phagocytosed by macrophages. A fibrinous exudate, fibrinoid necrosis and hyaline membrane formation were observed (A: HE stain, 40 $\times$; B: HE stain, 200 $\times$; C: HE stain, 100 $\times$; D: Masson trichrome stain, 100 $\times$).

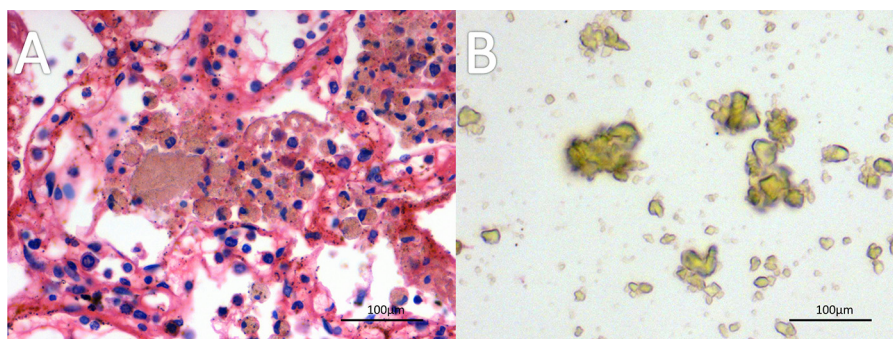


Fig. 5. Barium sulfate contrast material in lung (A) and direct smear (B) (A: HE stain, 400 $\times$; B: HE stain, 400 $\times$).

disturbance or motor neuron dysfunction [12]. Researchers consider it harmless to introduce barium sulfate into the bronchial tree as a bronchography contrast medium in practice [13,14]. A certain amount of barium aspiration does not cause symptomatic effects. However, fatal consequences can be incurred in individuals mentioned above. Considerable amounts of barium aspirated do obstruct major or peripheral bronchi, which reflect a bold outline of occluding air passage and the patients die from asphyxiation in a very short time [7]. When barium aspiration mixed with gastric contents, patchy depositions are significant in chest film. Death due to acute respiratory distress syndrome (ARDS) induced by severe chemical bronchitis and pneumonia can take place [8]. Apart from the aspirated volume, density of the suspension also contributes to the lung injury even that a small amount is well tolerated [15,16]. High density barium suspension causes the most damage and may be affected by the heterogeneity and size of the barium sulfate crystals [4,15,16]. Aspiration of large quantity, low-density barium sulfate is equally dangerous since the gas exchange will lead to mechanical interference [11,17]. In this patient, there might be a reflux without perception, possibly induced by volvulus and incomplete obstruction of ascending colon and

local anesthesia, which played a dominant role in the mixtures aspiration of the respiratory system. Microscopically, yellowish-brown foreign materials and intra-alveolar dense neutrophilic inflammatory infiltrates accompanied with fibrin exudates, hyaline membranes and fibrinoid necrosis of alveolar walls indicated that acute aspiration pneumonia and shock lung were definite. Herein, the cause of death should be chemical pneumonia rather than asphyxia even that a large dose of barium solution was aspirated.

Early warnings of chemical pneumonia can be dull and obscure, with few manifested tachypnea, cyanosis or wheezing rale due to bronchial spasm. Victoria et al. [18] considered that cyanosis, tachypnea, moist rale or tubular breath sound availably prompted the probability of chemical inflammation. If the principal clinical symptoms of barium aspiration are not typical, detailed history and radiographic examination play a pivotal part in inhalation pneumonia. In this case, lung reaction of the patient was seemingly mild in the early phase after the procedure. The attack of shock and pneumonia as cardiopulmonary anomaly occurred with a progressive drop in BP and rising in temperature 1 h post procedure. Foreign body aspiration manifested as patchy consolidation was confirmed by chest film 3 h later.

Besides aspiration, a wide spectrum of complications commonly appears in barium enema examination. The barium suspension and perfusion instruments may induce a mild-to-moderate anaphylactic reaction. In some cases, pulmonary fibrosis and interstitial nodular granuloma, even fistula, nodule and ulcer could be formed where some barium particles are deposited in a tissue erosion [19]. As previously reported, extravasation of barium sulfate into the peritoneal or retroperitoneal space resulted in a mortality of 75% in the aged individuals [12]. It also causes intestinal obstruction, fistula, abscess, and barium sulfate granuloma [20], which exhibits a higher risk in patients receiving a long term cortisol hormone therapy or having alimentary lesions. Intravasation into the blood stream is prone to embolize the corresponding segments [21] with a fatality rate of up to 55% [12]. A venous leakage of barium enema from upper rectal and colon may embolize the portal system whereas the leakage from lower rectal and vagina could cause acute pulmonary embolism via internal iliac vein return. Since the patient had SLE thus a medication history of prednisone, and incomplete volvulus of the colon ascendens was found in autopsy, we had checked carefully to exclude the possibility of extravasation and intravasation after systematic necropsy.

Even if the barium sulfate is stable and water-insoluble without being assimilated or metabolized in a natural alimentary tract, free barium ions in suspension can be dissolved in water and absorbed [22]. Seymour et al. [23] revealed that there was a significant statistical difference of barium contents detected from blood and urine several hours after a barium enema examination. In this backdrop, acute intoxication was induced because the digestive tract was abnormally permeable to the barium retention where there was intestinal or mucosal barrier obstruction. In 1999, Pélissier-Alicot et al. [21] reported an old woman presented symptoms characterized by non-specific neurologic and cardiovascular abnormality several hours after an oral administration of barium sulfate and died in 16 days. In her case, elevated levels of barium in her blood and cerebrospinal fluid were detected in clinical laboratory results. Given all that, whether the toxicity of barium sulfate contributed to the death is an inevitable question in this case. The main signs and symptoms of barium sulfate intoxication are dizziness, headache, numbness of the face and limb, and decreased muscle strength [24–26]. This patient merely manifested symptoms as respiratory embarrassment rather than any neurologic or mental disorder. The interval from symptom onset to mortality was 10 h. On necropsy, no signs pointed to complications related to alimentary of barium sulfate due to intravasation from blood vessel rupture or extravasation by leaking. Hence, both clinical manifestations and necropsy findings could exclude intoxication of barium sulfate.

A fatality related to lethal aspiration of barium sulfate is rare in both clinical and forensic medicine. Review of the medical records, radiographic results in particular, serves a guiding role in the determination of the cause of death. Severity of pulmonary inflammation and morphology of contrast medium could be observed via HE and Masson staining. And the underlying diseases, post-aspiration interval and clinical manifestations should be investigated exhaustively to take other contrast-related complications and circumstances into consideration.

Conflict of interest

None.

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